FCT Scientific Initiation Fellowship Report

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1 Introduction

This work aimed to study rearrangements for elliptic partial differential equations. In particular, the initial focus centered on the study of inequalities developed by Giorgio Talenti (1976) and their applications to a specific class of Dirichlet problems. We explored how Schwartz symmetrization allows obtaining *a priori* estimates for solutions, as well as the limitations of the main inequality in asymmetric configurations.

The natural corollary of this study led us to investigate shape optimization problems and source localization, culminating in the resolution of an open problem: the Tangency Conjecture for optimal source balls.

2 The Main Inequality: The Single Ball Case

The central inequality of this approach, which illustrates the power of Talenti's principle in an elementary setting with the indicator function, is given below. This case served as the theoretical foundation for the rest of the project.

Theorem 1 (Applied Main Inequality)**.** *Let $R > 0$, $a \in B_R(0)$ and $0 < \varepsilon < R - ||a||$ be fixed, and let u, v be solutions to the following Dirichlet problems:*

$$\begin{cases} \Delta u = \mathbf{1}_{B_\varepsilon(a)} & \text{in } B_R(0) \\ u = 0 & \text{on } \partial B_R(0) \end{cases} \quad \text{and} \quad \begin{cases} \Delta v = \mathbf{1}_{B_\varepsilon(0)} & \text{in } B_R(0) \\ v = 0 & \text{on } \partial B_R(0) \end{cases}.$$

Then, we have that the Schwartz symmetrization of u, denoted by u^*, satisfies

$$u^* \leq v$$

pointwise in $B_R(0)$.

Proof. Defining $U = -u$ and $V = -v$, we rewrite the problems for the positive Laplacian. Assuming $f(x) = \mathbf{1}_{B_\varepsilon(a)} \geq 0$, which represents no loss of generality, the Maximum Principle guarantees that the solution U is positive in the domain $G = B_R(0)$. We then have:

$$-\Delta U = f(x) \quad \text{and} \quad -\Delta V = f^*(x)$$

where f^* denotes the decreasing rearrangement of the source.

We begin by integrating both sides of the equation $-\Delta U = f(x)$ over the level set $\{x \in G : U(x) > t\}$. By Gauss's theorem, and denoting the gradient of U by DU, we obtain the following inequality:

$$\int_{U(x)=t} |DU| H_{n-1}(dx) \leq \int_{U(x)>t} f(x)dx \tag{1}$$

for almost every $t > 0$, where H_{n-1} represents the $(n-1)$-dimensional measure.

Considering the distribution function $\mu(t)$, the formula for its derivative is given by $-\mu'(t) = \int_{U(x)=t} \frac{1}{|DU|} H_{n-1}(dx)$. Using this formula together with Schwarz's inequality and the isoperimetric inequality $H_{n-1}\{x \in G : U(x) = t\} \geq nC_n^{1/n}\mu(t)^{1-1/n}$ (adapting the dimension m from the original article to n), we obtain the following lower estimate:

$$\int_{U(x)=t} |DU| H_{n-1}(dx) \geq n^2 C_n^{2/n} \mu(t)^{2-2/n} \frac{-1}{\mu'(t)} \tag{2}$$

for almost every $t > 0$, where C_n is the measure of the n-dimensional unit ball.

We now have to bound the right-hand side of equation (1) via the distribution function μ. Resorting to the rearrangement property, we have that $\int_{U(x)>t} f(x)dx \leq \int_0^{\mu(t)} f^*(s)ds$. In our specific case, f is the indicator of a ball of volume $|B_\varepsilon|$, so its decreasing rearrangement $f^*(s)$ takes the value 1 for $s < |B_\varepsilon|$ and 0 for the remaining values. Thus, this integral is exactly $\min(\mu(t), |B_\varepsilon|)$.

From (1), (2), and the previous estimate, we obtain the fundamental inequality:

$$1 \leq \frac{1}{n^2 C_n^{2/n}} (-\mu'(t)) \mu(t)^{-2+2/n} \int_0^{\mu(t)} f^*(s)ds \tag{3}$$

for almost every $t > 0$.

Note that the right-hand side of equation (3) is the derivative of an increasing function of t; integrating both sides between 0 and t, we obtain:

$$t \leq \frac{1}{n^2 C_n^{2/n}} \int_{\mu(t)}^{|G|} r^{-2+2/n} \left(\int_0^r f^*(s)ds \right) dr \tag{4}$$

.

By the definition of the decreasing rearrangement U^* of U, this inequality implies that:

$$U^*(s) \leq \frac{1}{n^2 C_n^{2/n}} \int_s^{|G|} r^{-2+2/n} \left(\int_0^{r'} f^*(s')ds' \right) dr' \tag{5}$$

.

Finally, through the definition of spherically symmetric rearrangements, the last inequality can be rewritten as follows:

$$U^*(x) \leq V(x) = \frac{1}{n^2 C_n^{2/n}} \int_{C_n|x|^n}^{|G|} r^{-2+2/n} \left(\int_0^{r'} f^*(s')ds' \right) dr' \tag{6}$$

achieving our goal, since V is exactly the solution of the symmetrized Dirichlet problem. Since $U^*(x) \leq V(x)$ and knowing that the original functions were strictly negative, translating back we conclude that $u^* \leq v$ pointwise.

Since $V(x)$ is a radial function ($V(x) = V(r)$ with $r = |x|$), we can deduce its exact form. For $n \geq 3$, the equation in polar coordinates is:

$$\frac{1}{r^{n-1}} \frac{d}{dr} \left(r^{n-1} \frac{dV}{dr} \right) = -\mathbf{1}_{[0,\varepsilon)}(r)$$

Integrating for $r < \varepsilon$, we obtain $\frac{dV}{dr} = -\frac{r}{n}$. For $r > \varepsilon$, as the source is zero, integration yields $\frac{dV}{dr} = -\frac{\varepsilon^n}{nr^{n-1}}$. Integrating from $r > \varepsilon$ up to R, and applying the boundary condition $V(R) = 0$, we obtain the expression for the tail of the potential:

$$V(r) = \frac{\varepsilon^n}{n(n-2)} \left(r^{2-n} - R^{2-n} \right)$$

□

3 The Resolution of the Tangency Conjecture

Supported by the principles of symmetrization and monotonicity explored in the previous section, the main focus of our subsequent investigation fell upon the optimal geometric configuration for *multiple* sources. In particular, we sought the arrangement of source balls that maximizes the L^∞ norm of the corresponding potential, thereby solving the so-called "Tangency Conjecture".

Consider the domain $\Omega = B_R(0) \subset \mathbb{R}^n$, with $n \geq 2$, and assume $0 < \epsilon < R$. For points $x, y \in \mathbb{R}^n$ satisfying $|x| \leq R - \epsilon$ and $|y| \leq R - \epsilon$, we denote by $u_{x,y}$ the solution to the Poisson problem:

$$\begin{cases} -\Delta u_{x,y} = \mathbf{1}_{B_\epsilon(x)} + \mathbf{1}_{B_\epsilon(y)} & \text{in } \Omega \\ u_{x,y} = 0 & \text{on } \partial\Omega \end{cases} \tag{7}$$

Let $\mathcal{A}$ be the set of admissible pairs (x, y) such that the balls do not overlap excessively ($|x-y| \geq 2\epsilon$). During this project, we successfully proved unequivocally the following central result:

Theorem 2 (Tangency of a Global Maximizer)**.** *Every global maximizer $(x, y) \in \mathcal{A}^*$ of the functional*

$$(x, y) \mapsto ||u_{x,y}||_{L^\infty(\Omega)}$$

strictly satisfies the condition $|x - y| = 2\epsilon$.

The detailed proof of this result is based on the directional monotonicity of the Green's function, the moving planes method, and geometric reflection.

4 Conclusion

The detailed proof for the single ball case (indicator and Laplacian) served as a rigorous basis for understanding symmetric rearrangements and their pointwise barriers. Most importantly, this geometric and analytical intuition gave rise to the resolution of the Tangency Conjecture, proving that two globally optimal sources in a spherical domain are, in fact, necessarily tangent.

References

[1] G. Talenti. *Elliptic equations and rearrangements*. Annali della Scuola Normale Superiore di Pisa, 1976.

[2] L. C. Evans. *Partial Differential Equations*. American Mathematical Society, 1998.